OXFORD AQA
INTERNATIONAL QUALIFICATIONS

INTERNATIONAL GCSE

PHYSICS

9203/1

Paper 1

Mark scheme

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:
- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	**Response**	**Marks awarded**
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	**Response**	**Marks awarded**
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of ‘it’

Answers using the word ‘it’ should be given credit only if it is clear that the ‘it’ refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…..) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	gravitational force		1	AO1 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	helium hydrogen		1 1	AO1 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	density		1	AO1 3.8.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	main sequence		1	AO1 3.8.1b,j

<table>
<tr><th>Question</th><th colspan="2">Answers</th><th>Extra information</th><th>Mark</th><th>AO/
Spec. Ref.</th></tr>
<tr><td rowspan="2">01.5</td><td colspan="3">all ticks correct

allow 1 mark for two or three correct ticks</td><td rowspan="2">2</td><td rowspan="2">AO1
3.8.1j</td></tr>
<tr><td colspan="3">
<table>
<tr><th>Stage of the life cycle</th><th>Stars about the same size as the Sun</th><th>Stars much bigger than the Sun</th></tr>
<tr><td>Red giant</td><td>✓</td><td></td></tr>
<tr><td>Red supergiant</td><td></td><td>✓</td></tr>
<tr><td>White dwarf</td><td>✓</td><td></td></tr>
<tr><td>Black dwarf</td><td>✓</td><td></td></tr>
<tr><td>Neutron star</td><td></td><td>✓</td></tr>
<tr><td>Black hole</td><td></td><td>✓</td></tr>
</table>
</td></tr>
</table>

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	supernova		1	AO1 3.8.1i

Total Question 1		**8**

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	$W = 7.5 \times 9.8$		1	AO2 3.1.1e
	$W = 73.5$ (N)	allow 74 (N)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	attach masses / weights to the spring to apply a force (of 150 N)	accept attach the newtonmeter to the spring and extend until the newtonmeter reads 150 N	1	AO4 3.1.1h
	measure the length of the spring with a ruler		1	
	then subtract the initial length from the final length		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	$150 = 400 \times e$		1	AO2 3.1.1h
	$e = \frac{150}{400}$		1	
	$e = 0.375$ (m)	allow 0.38 (m)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	$E = \frac{1}{2} \times 400 \times 0.16^2$		1	AO2 3.2.1c
	$E = 5.12$ (J)	allow 5.1 (J)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	• gravitational potential energy of the baby decreases		1	AO2 3.2.2a
	• kinetic energy of the baby increases (then decreases)		1	
	• elastic potential energy of the spring increases	if no other mark awarded allow **1** mark for energy is dissipated to the surroundings	1	

Total Question 2		**13**

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	vector quantities have (magnitude and) direction		1	AO1 3.1.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	average speed = $\frac{400}{80}$		1	AO2 3.1.2c,d
	average speed = 5 (m/s)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	330 = $m \times 5.5$		1	AO2 3.1.4a
	$m = \frac{330}{5.5}$		1	
	m = 60 (kg)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	displacement is a vector quantity	allow a description of displacement	1	AO2 × 1 AO3 × 1 3.1.1d
	(start and) end of the race is in the same place		1	

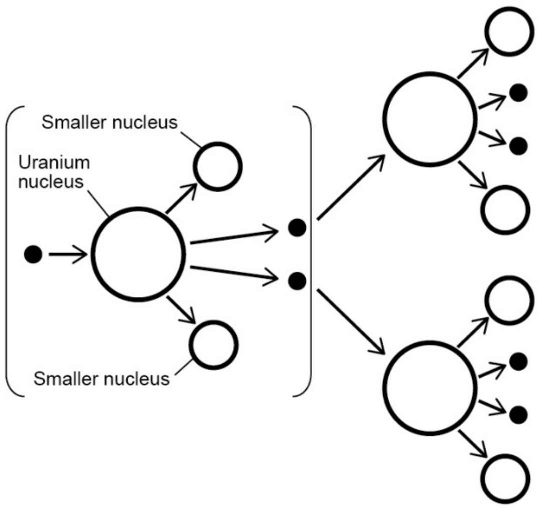
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	decelerating	allow speed is decreasing	1	AO2 3.1.3d
	at (a rate of) 1 m/s each second		1	

Total Question 3		**10**

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	uranium-235		1	AO1 3.7.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	1	this order only	1	AO1 3.7.1c
	0		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	neutron aimed at a new uranium nucleus	accept neutron close to a new uranium nucleus without an arrow shown	1	AO1 3.7.3d
				
	2 smaller product nuclei and 2 / 3 neutrons shown		1	
		note: either one or two of the fission neutrons could be the start of the chain reaction		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	absorb (excess fission) neutrons		1	AO1 3.7.3d
	(on average only) one fission neutron per reaction produces one further fission reaction		1	
	keeps the rate of reactions constant		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	drop them fully into the reactor		1	AO4 3.7.3d

<table>
<tr><th>Question</th><th>Answers</th><th>Mark</th><th>AO/ Spec. Ref.</th></tr>
<tr><td rowspan="4">04.6</td><td>Level 2: Scientifically relevant features are identified; the way(s) in which they are similar / different is made clear and the magnitude of the similarity / difference is noted.</td><td>3–4</td><td rowspan="4">AO1 × 2
AO3 × 2
3.2.3 a
3.7.3 c,e</td></tr>
<tr><td>Level 1: Relevant features are identified and differences noted.</td><td>1–2</td></tr>
<tr><td>No relevant content</td><td>0</td></tr>
<tr><td>Indicative content

• oil and coal generate less energy (per kg) than uranium
• (burning) oil and coal produces carbon dioxide
• (burning) oil and coal contribute to global warming or climate change

• (nuclear fission) produces radioactive / nuclear waste
• nuclear power stations are expensive to decommission
• in the event of an accident, nuclear power stations cause more widespread damage to the environment than oil or coal power stations
• nuclear fission of uranium does not produce carbon dioxide
• none of the fuels are renewable

• uranium generates 2 million times more energy per kg than oil
• coal generates 1.5 times more energy (per kg) than oil
• uranium generates over 1.3 million times more energy (per kg) than coal

for Level 2 answers must include information not given in Table 2</td><td></td></tr>
</table>

Total Question 4		**13**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	evaporation		1	AO1 3.4.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	any **two** from: • temperature of air • air flow • humidity • surface area of clothes • how wet the clothes are at the start • material of clothes	 allow amount of sunlight allow wind speed	2	AO1 3.4.2b

<table>
<tr><th>Question</th><th>Answers</th><th>Mark</th><th>AO/
Spec. Ref.</th></tr>
<tr><td rowspan="5">05.3</td><td>Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.</td><td>5–6</td><td rowspan="5">AO2 × 2
AO3 × 4
3.6.5f</td></tr>
<tr><td>Level 2: Some logically linked reasons are given. There may also be a simple judgement.</td><td>3–4</td></tr>
<tr><td>Level 1: Relevant points are made. They are not logically linked.</td><td>1–2</td></tr>
<tr><td>No relevant content</td><td>0</td></tr>
<tr><td>Indicative content

• power of device B is less
• drying time for device B is greater
• cost of device B is more than device A

• power (of device B) 20 times smaller
• (but drying time of device B) 8 times greater
• device B uses less energy / is cheaper to run (per load)
• cost of device B is (more than) 2 times device A

• (energy per load for device A) 3 (kW) $\times$ 0.75 (h) = 2.25 (kWh)
• (energy per load for device B) 0.150 (kW) $\times$ 6(h) = 0.9 (kWh)
• 2.25 (kWh) > 0.9 (kWh)
• $\frac{(\$)510}{(\$)250}$ = 2.04 times the price of device A

for Level 2 answers must include a calculation

for Level 3 answers must include reference to power, time and cost</td><td></td></tr>
</table>

Total Question 5		9

Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	iris		1	AO1 3.3.6h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	cornea		1	AO1 3.3.6h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	$1.44 = \frac{300\ 000\ 000}{v}$		1	AO2 3.3.5d
	$v = \frac{300\ 000\ 000}{1.44}$		1	
	v = 208 333 333 (m/s)		1	
	$v = 2.08 \times 10^8$ (m/s)	allow answer in standard form to 2 or more significant figures	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	$0.0012 = \frac{0.0055}{\text{object height}}$	allow incorrect / not converted value of *image height* for subsequent marks	1	AO2 3.3.6g
	object height $= \frac{0.0055}{0.0012}$		1	
	object height = 4.583333333 (m)		1	
	object height = 4.6 (m)		1	

Total Question 6		**10**

Question 7

<table>
<tr><th>Question</th><th>Answers</th><th>Extra information</th><th>Mark</th><th>AO/
Spec. Ref.</th></tr>
<tr><td>07.1</td><td>
<table>
<tr><th>Use</th><th>Electromagnetic wave</th></tr>
<tr><td>Bluetooth communication</td><td>Radio waves</td></tr>
<tr><td>Medical imaging</td><td>X-rays</td></tr>
<tr><td>Night vision devices</td><td>Infrared</td></tr>
<tr><td>Security marking</td><td>Ultraviolet</td></tr>
<tr><td>Sterilising surgical equipment</td><td>Gamma rays</td></tr>
</table>
</td><td>1 mark for each correct answer</td><td>2</td><td>AO1
3.3.2h</td></tr>
</table>

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	X-rays are ionising so can kill / damage cells	allow can cause cancer	1	AO1 × 2 AO2 × 1 3.3.2i
	so they are dangerous for the doctor if used many times		1	
	the lead screen absorbs X-rays (reducing the amount of ionising radiation reaching the doctor)	allow lead screen reduces dose (to doctor)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	the metal artificial hip absorbs the most X-rays so appears white (on image)	accept metal artificial hip transmits very little X-rays so appears white (on image)	1	AO3 3.3.2k
	normal hip bone absorbs some X-rays so appears light	accept normal hip bone does not transmit many X-rays so appears light	1	
	muscle transmits most X-rays so appears dark	accept muscle does not absorb many X-rays so appears dark	1	

Total Question 7		**8**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	dc is always in the same direction		1	AO1 3.6.3a,b
	ac is repeatedly changing direction		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	1 A		1	AO2 3.6.5c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	the Earth wire provides a low resistance path for the current		1	AO1 3.6.3f
	(the large current) melts the fuse		1	
	(and) disconnects the live wire	allow breaks the circuit	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	these devices can't become live		1	AO1 3.6.3f
	because they have a plastic casing	allow insulating material for plastic allow because they are double insulated	1	

Total Question 8		**8**

Question 9

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.1	2 (minutes)		1	AO3 3.4.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.2	$\Delta\theta = 88 - 69$		1	AO3 3.4.1e
	$\Delta\theta = 19$ (ºC)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.3	t = (22 – 6 = 16 min =) 960 s	allow 22 to 22.5 and/or 5.5 to 6 allow correct use of their value of t for subsequent marks	1	AO2 3.4.1d 3.2.1f
	$2.4 = \frac{E}{960}$		1	
	$E = 2.4 \times 960 = 2304$ J		1	
	$2304 = m \times 2.0 \times 10^5$		1	
	$m = \frac{2304}{2.0 \times 10^5}$		1	
	$m = 0.01152$ (kg)	allow 0.012 or 0.0115 maximum **3** marks if t is not taken from horizontal part of the graph	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
09.4	line starts at 88 °C and has lower gradient		1	AO3 3.4.1e
	horizontal part of the graph approximately the same length and at 69 ºC	allow if line is longer	1	

Total Question 9		**11**